

Module 07: Decision Support Systems in the Water Domain

Decision Support Systems Class Exercise and Assignment definition

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The purpose of these exercises is to introduce you to the use of Multi Criteria Analysis methods using a python library (pymcdm), developed to assist in applying different MC methods.

The PYMCDM Library requires installation of a python environment on your Computer.

Part I: Installing Python and creating environments on your computer

In order to use the Multicriteria analysis library, you will need to install a Python environment on your own PC. There are several options available, which are described below. Use Option 1 or Option 2 to install the Python environment.

Option 1. Installing a full Python environment on your PC

Step 1: Installing Python in Windows 11

In order to install a python environment, you have several options:

- Anaconda (Recommended)
- Miniconda and VS code

Instructions are available as follows:

- For Python any of these websites are good installation guides:
 - <https://phoenixnap.com/kb/how-to-install-python-3-windows>
 - <https://learn.microsoft.com/en-us/windows/python/beginners#install-python>
 - <https://learn.microsoft.com/en-us/windows/python/beginners#install-python>
- Anaconda: download and install: <https://www.anaconda.com/>

We recommend installing Anaconda3, and use of Spyder (version 6) or Jupyter notebooks from the Anaconda Installation.

We will be using a specialized python library for this course called `pymcdm`. All information about the library is available at: <https://pypi.org/project/pymcdm/> or <https://pymcdm.readthedocs.io/en/master/>

All class examples are done with version 1.2.1.

Step 2: Create a dedicated conda environment for the class exercises

To get started, open the Anaconda Prompt and type the following command to create a conda environment called `dss-course`. Notice that we are specifying the version of Python for this environment.

```
conda create -n dss-course python=3.11.9
```

Now, activate the environment that was just created and install the necessary packages.

```
conda activate dss-course
conda install pandas
pip install pymcdm==1.2.1
```

If you are using **Spyder** as your code editor, then you need to install the appropriate spyder kernels depending on your version of Spyder:

For Spyder 5:

```
pip install spyder-kernels==2.5.*
```

For Spyder 6:

```
pip install spyder-kernels==3.0.*
```

Step 3. Replace a set of files on your computer with the ones available on eCampusXL

First, you need to locate the `dss-course` conda environment on your computer. To do this, run the following in the Anaconda command prompt:

```
conda info --envs
```

This will display the paths to all your conda environments. The output will look like something like this:

```
base          C:\Users\mca010\miniconda3
dss-course    * C:\Users\mca010\miniconda3\envs\dss-course
```

Find the path next to `dss-course` and copy it. Paste this path into the address bar of your file browser, then:

- Navigate to the `Lib` folder
- Then go to `site-packages`
- Locate the `pymcdm` folder

Extract the files from “ReplaceFiles.zip”, available on eCampus.

Copy extracted files on your computer on the folder where you have the `pymcdm` installed.

IMPORTANT: Do not copy the “methods” and “visual” folders from ReplaceFiles over your Anaconda installation, just add files from those folders in the respective files in Anaconda. Accept replacement of files, i.e. overwrite those particular existing files on your computer.

Step 4. Configure Spyder with the correct conda environment

To set Spyder to use the correct conda environment:

- Open Spyder.
- Go to **Tools** → **Preferences** in the menu bar.
- Click the **Python interpreter** tab.
- Select **Use the following interpreter:**
- Paste the full path to your conda environment, including `python.exe` at the end.

For example:

```
C:\Users\<your-user-name>\AppData\Local\anaconda3\envs\dss-
course\python.exe
```

Before continuing with the tutorial, you need to restart the Spyder Kernel. To do this go to the menu bar of Spyder and click **Consoles** → **Restart kernel**.

If the above does not work: Go to **File** → **Restart**, in order to completely restart Spyder.

Option 2. Installing an isolated Python environment

This option guide the set- up of an isolated Python environment for the class exercises using `uv`, a fast, modern Python package manager. Nothing is installed globally on your PC.

Step 1. Prerequisites

Before starting, make sure you have the following:

- Windows 11 (64-bit)
- Spyder 6 installed
- We recommend a stand-alone installation (<https://www.spyder-ide.org/>), but you can also use the one that comes with your Anaconda installation (<https://www.anaconda.com/download/>)
- An internet connection for the initial `uv` installation
- The class exercises zip file “ExercisesDSS-uv.zip” from eCampusXL

Step 2. Install `uv`

`uv` is a single executable. The installer downloads it automatically and adds it to your PATH:

- *Run the installer*

Open PowerShell (search for it in the Start menu) and paste the following command:

```
powershell -ExecutionPolicy ByPass -c "irm  
https://astral.sh/uv/install.ps1 | iex"
```

Wait for the installer to finish. You should see a confirmation message.

- *Verify the installation*

Close PowerShell and open a new PowerShell window. Then run:

```
uv --version
```

You should see a version number printed, for example:

```
uv 0.5.x (or higher)
```

If `uv` is not found after reopening PowerShell

Close and reopen PowerShell. If the error persists, restart your computer. The PATH update sometimes requires a full restart on Windows 11.

Step 3. Create the Python Environment

- *Extract the zip file*

Locate the tutorial zip file. Right-click on it and select Extract All. Choose a location you can easily navigate to, for example:

```
C:\Users\YourName\Documents\ExercisesDSS
```

Note: Avoid paths with spaces or special characters (e.g. accented letters). This can cause issues with some Python packages.

- *Navigate to the folder for the exercises*

In PowerShell, navigate to the folder you just extracted. Replace the path below with your actual path:

```
cd C:\Users\YourName\Documents\ExercisesDSS
```

- *Run uv sync*

This single command creates the environment and installs all packages, including the modified tutorial package:

```
uv sync
```

uv will read pyproject.toml and install everything automatically. The first run may take 30-60 seconds. Subsequent runs are near-instant.

When it finishes you will see output similar to:

```
Resolved 98 packages in 33ms
Installed 16 packages in 3.4s
+ numpy==2.x.x
+ pandas==2.x.x
+ matplotlib==3.x.x
+ pymcdm==1.2.1+ipo
...
```

Step 4. Connect the Environment to Spyder

- *Install Spyder Kernel*

For using Spyder, then the appropriate spyder kernels need to be installed. The version of the package to be installed will be depending on your version of Spyder by using the following commands in Powershell:

For Spyder 5:

```
uv add spyder-kernels==2.5.*
```

For Spyder 6:

```
uv add spyder-kernels==3.0.*
```

- *Find the Python interpreter path*

Spyder needs to be told to use this environment instead of its built-in Python. You only need to do this once. In PowerShell, while still inside the tutorial folder, run:

```
uv run python -c "import sys; print(sys.executable)"
```

This will print a path like:

```
C:\Users\YourName\Documents\ExercisesDSS\.venv\Scripts\python.e
```

```
xe
```

Copy this path — you will paste it into Spyder in the next step.

- *Set the interpreter in Spyder*

Open Spyder, then follow these steps:

1. Go to Tools in the top menu bar
2. Select Preferences
3. In the left panel, click Python interpreter
4. Select Use the following Python interpreter
5. Paste the path you copied in the previous step
6. Click OK

- *Restart the Spyder kernel*

In the IPython Console panel (bottom right), click the red square (Stop) button, then the play button (Restart kernel). Alternatively, go to:

```
Consoles -> Restart kernel
```

The console will restart and now use the tutorial environment.

Step5. Verify Everything Works

In the Spyder IPython console, run the following to confirm the environment is active:

```
import sys
print(sys.executable)    # should show the .venv path

import numpy
print(numpy.__version__)

import pymcdm
print("Environment ready!")
```

If all three imports succeed without errors, your environment is correctly configured and you are ready to begin the exercises.

If you see a `ModuleNotFoundError`

Double-check that the Python path in Spyder Preferences matches exactly what `uv run python -c "import sys; print(sys.executable)"` printed, and that you restarted the kernel after changing the preference.

Part II Exercises

2.1. Use of the template file to apply a mcdm method

A template file on how to use the library for your analysis is provided on eCampus. The file will be used during the exercise sessions to solve problems. The *data.csv* example file is also available for download on eCampus.

During exercise sessions you will need to create your own data input file. The one provided on eCampus is just an example.

The Python code included in the template is

```
23# Define decision matrix (xx alternatives, yy criteria) by reading a csv file
24
25df = pd.read_csv('data.csv')
26# Use only columns with numerical data
27alts = df[df.columns[1:]].to_numpy()
28
29#to see the data in the csv file uncomment the next line
30#print(df)
31
32# Define weights and types
33#weights = w.equal_weights(alts)
34weights = np.array([0.33, 0.2, 0.13, 0.07, 0.27])
35types = np.array([1, -1, 1, 1, 1]) # cost or benefit attributes
36
37methods = [
38    TOPSIS(),
39    SAW()
40]
41
42method_names = ['TOPSIS', 'SAW']
43# Determine preferences and ranking for alternatives
44prefs = []
45ranks = []
46
47for method in methods:
48    pref = method(alts, weights, types)
49    rank = rrankdata(pref)
50
51    prefs.append(pref)
52    ranks.append(rank)
53
54# printing results
55a = [f'A{{{i+1}}}' for i in range(len(prefs[0]))]
56print('Preference table/ method \n')
57print(pd.DataFrame(zip(*prefs), columns=method_names, index=a).round(2))
58print('\n Ranking of alternatives / method \n')
59print(pd.DataFrame(zip(*ranks), columns=method_names, index=a).astype('int'))
60
61#different figures
62fig, ax = plt.subplots(figsize=(7, 3), dpi=150, tight_layout=True)
63visuals.ranking_bar(ranks, labels=method_names, ax=ax)
64plt.show()
65
66fig, ax = plt.subplots(figsize=(7, 7), dpi=150, tight_layout=True, subplot_kw=dict(proje
67visuals.polar_plot(ranks, labels=method_names, legend_ncol=2, ax=ax)
68plt.show()
```

Running the code will give results as follows:

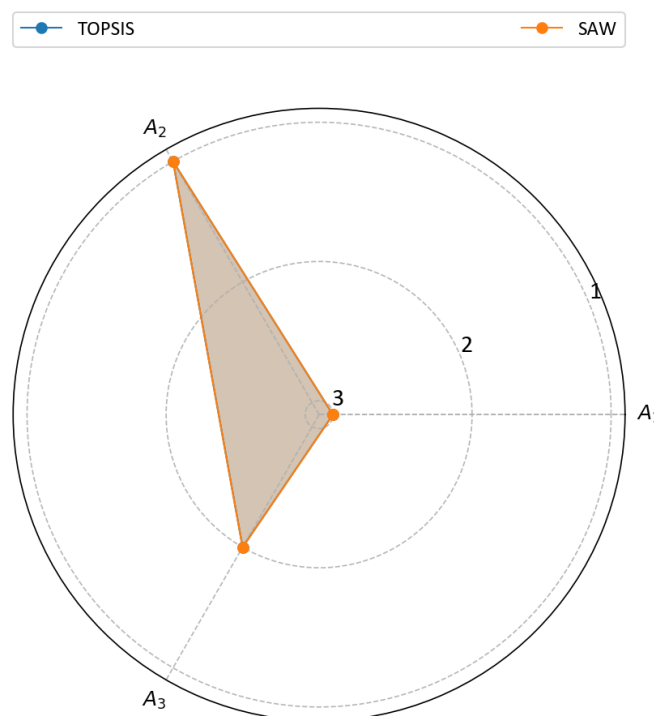
Preference table/ method

	TOPSIS	SAW
A{1}	0.35	0.48
A{2}	0.74	0.72
A{3}	0.47	0.54

Ranking of alternatives / method

	TOPSIS	SAW
A{1}	3	3
A{2}	1	1
A{3}	2	2

Example of a visual graph for the solution is:

*Note on programming*

The provided template file does not require you to know programming, it will be used as is. However, if you have experience in programming you can use other functions from the library, or improve the template, by adding your own lines of code.

2.2 Example problem to demonstrate the use of the python template

Problem definition: A town wants to expand its water supply from groundwater sources. Three potential well field locations are identified. The following are known about the well fields:

- Each well field yields different discharge Q . The more discharge is available the better.
- Installation and operation of these wells have different costs C
- The region where the well fields are to be located is a mixture of forests and wetlands. The impact of pumping is lowering of the groundwater levels, which result in reduction of wetland area and negative impact on some of the trees in the forest.
- The attitude of the citizens with regards to the installation of the wells is different for each of the three locations

Solution: For this given problem, five criteria for evaluation of alternatives are considered:

1. The discharge Q – estimated in m^3/s (benefit)
2. The cost C – estimated in euros (cost)
3. The impact on the wetlands, as loss of wetland area – estimated in ha (cost)
4. The impact on the forest, as average drawdown of groundwater table – estimated in meters (cost)
5. The social acceptance to any of the three potential locations, assessed qualitatively by using linguistic terms (high, low, medium) (benefit)

The value of these attributes is given in table below:

	Discharge (Q)	Cost (C)	Wetland (W)	Forrest (F)	Social acceptance
Alternatives (well loctions)	10^6 $m^3/year$	1000 euros	ha	meters	/
A1	2.5	50	500	0.7	low
A2	3	60	850	0.2	high
A3	4	80	1000	0.3	medium
	max	min	min	min	max

Implement first the case using an excel approach, then use the Python template code to see the result.

2.3. Pilot study application in the Venice Lagoon catchment

This exercise is a pilot study application addressing an irrigation problem in the Destra Piave region, in the Venice region.

Problem description:

Venice Lagoon and its watershed form a vast environmental system of approximately $2500 km^2$, where historical and recent cities, large and medium-small industrial districts and intensive agricultural activities coexist within a peculiar natural area. The drainage basin consists of a low gradient floodplain with a surface of about $2000 km^2$. It is drained by water courses with different hydraulic regimes (natural, mechanical, alternate mechanical). About 40% of the basin surface is below the mean sea level and is under reclamation by pumping machines. The

pollution discharged into the Venice Lagoon by the rivers streams and canals flowing through the above described catchment represents a relevant contribution to the overall pollution budget of the Lagoon and is a main environmental issue.

In Italy public funds are made available by specific national and regional regulations in order to support the realisation of initiatives for the abatement of pollutant loads that travel from the catchment into the lagoon. In particular, the Special Law for Venice (L. 171/73) defines measures aimed at protecting the city and its lagoon and confers the allocation of funds to the Regional Administration. To apply for those funds, local agencies in charge of water management can present suitable projects targeted at reducing diffuse pollution in the drainage basin.

One of the recipients of such funds are the Land Reclamation Boards, public administrative bodies made up of all the private individuals who live or have economic activities in a given territory requiring water management (defined according to hydrological and administrative criteria). Their main task is that of managing the water distribution system, protecting the territory from floods, managing and maintaining the public infrastructure for land reclamation and irrigation.

The Piave Destra irrigation system

The problem to be solved in the Destra Piave irrigation system is related to the management of the irrigation system itself. The current irrigation system is characterised by scarce efficiency in water use:

1. The existing distribution network is very old and requires substantial interventions
2. Land concerned is not particularly suitable for surface irrigation system
3. System operation and maintenance is very labour-intensiveThe increasing urbanisation process leads to increasing management problems



The Piavre Destra region

Solution: The solution to reduce the amount of water used for irrigation is to increase the water efficiency by using sprinklers for irrigation. Use a python code, based on the data provided in Annex 1 to choose the best option scenario.

Part III – Groupwork definition

Present exercise is adapted from a case study kindly provided by Dr. Abdulkarim H. Seid, from Nile Basin Initiative.

3.1. Case study description

A River has its source at the mountain tops and flows through steep slopes and narrow gorges. It terminates at its confluence with the ocean after supplying water for irrigation, replenish a wetland and it is used for navigation. The entire catchment is divided in two parts: **Upper Basin** and **Lower Basin** (see Figure 1 below)

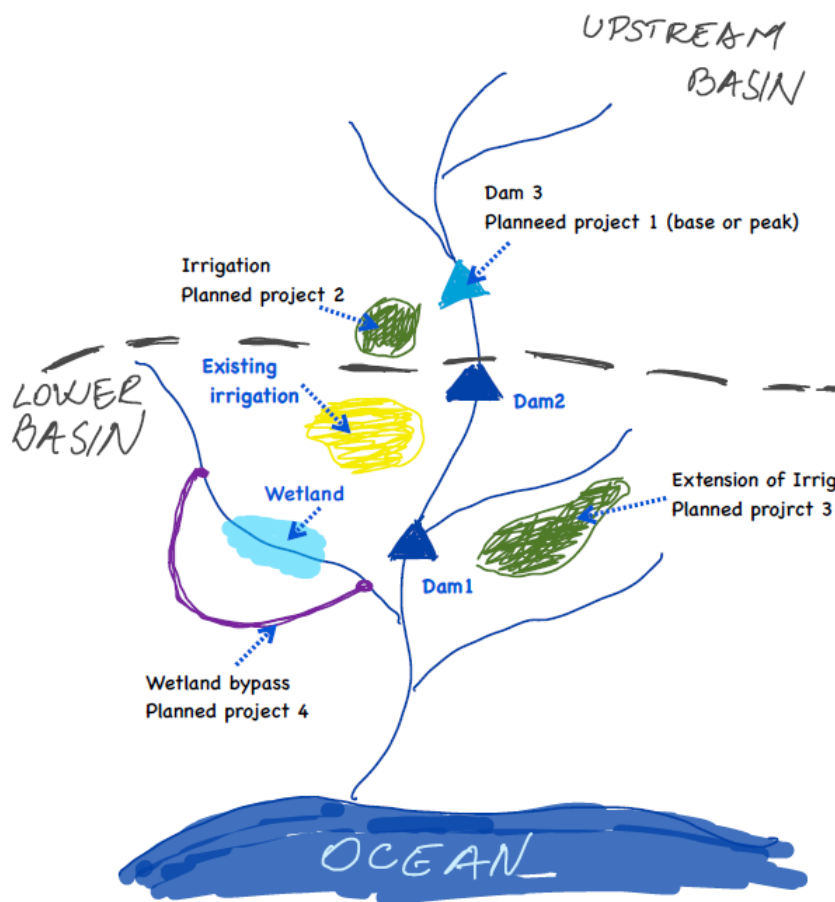


Figure 1. River basin: Upper and Lower Basin future and existing stakeholders

The **Upper Basin** makes up 24% of the total area and contributes 70% of the water that flows into the main River. The catchment here is steep leading to the problem of extensive soil erosion. The river and its tributaries are confined by narrow and deep gorges. The extreme seasonal variations in flow and the presence of rapids and waterfalls in this river have made the upstream reach of the river unfavourable for navigation. However the topographical conditions favour hydropower production if the water flow is regulated by a dam.

The **Lower Basin** is characterized by arid type of climate where the rainfall amount is relatively small as compared to the upper catchment. This part of the basin is endowed with fertile land and levelled topography that is suitable for irrigation development. Irrigation can be developed for crop production in this area at a cheaper cost than in any part of the basin. The seasonality of the river flow however did not allow utilizing the irrigation potential of the area to its full capacity.

Further downstream in the drainage basin, there is a protected wetland. The wetland is the habitat for birds and rare fish species as well as it is the source of revenue from eco-tourism. During the wet season, the entire wetland is flooded and it is impossible to identify the main river course. Downstream of the wetland, the river depth does not vary much which creates favourable condition for navigation.

Navigation plays a great role for the local economy in the Lower Basin. The river reach is used for: (i) navigation to connect small cities, (ii) tourism and (iii) transfer of goods from the Lower Basin to the river terminal point, at the ocean with large ships. During the times of low flow however, the navigable reach of the River is reduced substantially and allows only smaller boats to navigate.

The increase in population and urbanization in the Upper Basin led to increased demand for electricity and food. This required the Upstream Basin authorities to plan for development of hydropower in the narrow gorges and develop irrigation in the Lower Basin. These development interventions will have effects on the wetland and navigation in the Lower Basin.

The Upper Basin and Lower Basin stakeholders agreed to develop basin wide model to study the possible benefits and negative impacts of the planned interventions to select the best future development plan.

A set of alternatives has been agreed upon, formed by combinations of several identified development projects. There are 13 alternatives, named from A1 to A13. Three of the alternatives (the one where the bypass project is implemented) have different amounts of water considered, and because of that you will see numbers such as A6.1, A6.2, etc. All 13 alternatives will be evaluated based on criteria expressed by economic, social and environmental indicators. In case of alternative A6, A7 and A11, only one choice will be considered. You will select the choice yourself.

The best alternative will be considered for implementation. The evaluation process will be supported by the Multi Criteria Analysis implemented in a python code.

3.2. Baseline condition

Currently no major water infrastructure exists in the Upper Basin, while there are irrigation schemes and two storage dams already in place in the Lower Basin (see figure 1)

Current conditions were evaluated by a group of stakeholders along with a series of identified indicators. Values of these indicators are given in the attached excel spreadsheet.

3.3. Planned developments

The key planned development projects are as follows (see Figure 1 for details):

Upper Basin:

- A hydropower plant drawing water from a storage dam ("Dam 3") – (code **1b** or **1p**, corresponding to base and peak energy production, respectively)
- New irrigated agriculture with water being supplied from the storage dam above; ("NIRR") – (code **2**)

Lower Basin:

- Irrigation expansion (“NIRR2”) – (code **3**)
- Bypass canal (“BYWET”) to improve navigability of river reach downstream (code **4**).
Different water volumes are considered for bypassing.

Key decisions the river basin authorities need to make are**Upper basin:**

Operation of the power plant: though the Upper Basin authorities decided to go ahead with the construction of the hydropower plant, it has not been decided yet whether to run the plant as base load or peak load plant.

Irrigated agriculture: though the primary objective of the storage dam is to meet hydropower production requirement, there is an opportunity to use the water for food production, i.e. through irrigation. Due to constraints in financial resources, the construction of the irrigation scheme might not materialize in the coming few years.

Lower basin:

Construction of the bypass canal: the primary objective of the bypass canal is to improve navigability of the river reach. Big opposition is expected from environmental lobby groups to the full scale development of the canal.

With the planned dam in the Upper basin, there is a big opportunity for improving the navigability of the river reach. However, if the upstream authorities decides to operate its new hydropower plant as peaking load plant, the improvement in the navigability would not be too high. In this regard, persuading the upstream stakeholders to run the hydropower plant as base load plant is important.

Irrigation expansion 2: the decision whether or not to go ahead with the planned irrigation expansion is one of the key decisions the Lower Basin. This planned expansion heavily depends on increased and sustained flow in the river; the planned storage dam upstream is expected to increase the river regulation.

The two river basins have recently formed a joint working group to promote cooperation on water resources management and agreed on sets of future development/management alternatives. These alternatives are defined in excel file “*AssignmentDefinedIndicators&Alternatives.xlsx*”, available on eCampusXL.

A set of 10 indicators was defined. All indicators values were computed using MikeHydroBasin. Their values, for each defined alternative, is provided in the same excel file. You will discuss and select the indicators based on which evaluation of alternatives will be done. The number of considered indicators should **not be less than 5**.

Assignment: Groupwork

You will work in groups. Each group will be formed from 2 subgroups; one representing the Upper Basin and one representing the Lower Basin. All alternatives will be evaluated jointly. Collaborate and negotiate in order to propose a final alternative to be implemented.

You will need to:

- Rank the scenarios and provide your joint recommendation on the best alternative that shall be considered for development
- Provide recommendation on any follow-on analysis/study you think should be conducted
- Evaluate alternatives using SAW, TOPSIS, and a third method of your choice. Explain the method quickly during your presentation.
- Use weighting coefficients for three p values: 0; 0.5; and 1
- Prepare a presentation of max 15 minutes, explaining your work. Presentation will be shown on 10th April, 2026, during the scheduled session. You need to upload the presentation in the evening before the session, i.e on 9th of April 2026.

Important remarks: Your final mark will be composed of an average between the mark for the groupwork and an individual mark, composed from your peers.

Logistics

In total there are 5 working groups; 3 groups of 7 students, and 2 groups of 8 students. Each group is split in two, half representing the Upper Basin, and half the Lower Basin. You work and evaluate all alternatives as a group, but negotiate based on the sub-basin (sub-groups) you represent.

You form the subgroup (i.e Upper basin/Lower Basin) yourself.

There are special requirements on how to form the groups, Please check them on the ppt that is explained in class.

Annexes

(a) Destra Piave data

The list of indicators, for the Destra Piave irrigation system, as they have been identified by the group of decision makers, is:

Sector		Indicator name	Description	Measurm ent unit	DPSI	Env Soc Ec
AGRICULTURE	1	CAP Reform	Effects of Fishler Reform on local agriculture	ND	D	011
	2	Impacts on farms and the rural community	Effects of measures on local farms	ND	I	011
	3	Land use and crop choice	Crop water requirements	m3/ha/yr	P	101
	4	Impacts of cultivation on soil	Effects of cultivation practices	ND	S	101
	5	Efficiency of irrigation water	Overall efficiency	%	P	101
WATER RESOURCES MANAGEMENT	1	Water Framework Directive implementation"	Effects of water policy on local agriculture	ND	D	111
	2	Social conflicts between social and environmental sectors	Conflicts deriving from limited water availability	ND	I	010
	3	Implementation and maintenance costs	Costs of the new irrigation system	€	I	001
	4	Synergies with other projects and activities	Possible civil use of irrigation water through a dual distribution system	m3/ha/yr	I	011
	5	Agreements about the use of the Piave River	Water rights	ND	D	111
URBAN LAND USE	1	Civil settlements	Typologies of settlements and related water uses	m3/inh/y r	D	010
	2	Water use technologies for civil uses"	Average consumption per technology	m3/inh/y r	P	010
WATER RESOURCES	1	Availability of water for environmental uses	Available water volumes	m3/ha/yr	S	101
	2	Impact on surface and ground-water resources	Water available for aquifer recharge and surface flow	m3/yr	I	100
	3	Pollution of surface and ground-water	Pollutants loads	kg/ha/yr	I	100
	4	Surface water flow	Water volumes in the canal network	m3/ha/yr	P	101
	5	Availability of water for farms	Adequate availability of water in quantity and time	m3/ha/yr	S	101

The values of the indicators for different scenarios are provided in an excel spreadsheet on eCampusXL.

(b) Assignment data is provided in an excel spreadsheet on eCampusXL.

Explanation of the data included in the excel spreadsheet will be given in class.